



SUMULONG MEMORIAL HIGH SCHOOL
Gen. Luna St. Brgy. San Roque, Antipolo City
SY. 2022-2023

CIRCULAR MOTION

LEARNING MODULE

NAME: _____ **YEAR AND SECTION :** _____



LEARNING OBJECTIVES:

At the end of this module, you are expected to:

1. Relate the laws of motion to bodies in uniform circular motion;
2. Infer that circular motion requires the application of constant force directed toward the center of the circle;
3. Solving the problems involving Newton's laws of motion.
4. Study the characteristics of uniform circular motion
5. Experimentally measure centripetal force in a circular motion.

OBJECTIVES:

1. Relate the laws of motion to bodies in uniform circular motion;
2. Infer that circular motion requires the application of constant force directed toward the center of the circle;

MATERIALS:

1. Hanger
2. Penny

**Try This
Inquiry Lab**

What Kept a Penny in the Coat Hanger?

INSTRUCTIONS

1. Put the middle of the longest straight part of the hanger out and shape it into a long narrow shape.
2. Let the hanger hang on your right index finger and place a penny on the end of the wire.
3. Start to swing the hanger –first slowly back and forth, then do full loops.
4. Swing it slower and try to catch the penny. Would the coin still to the hanger if the swing was done in a much slower pace.



LEARN MORE!

CENTRIPETAL FORCE

According to Newton's second law of motion, acceleration is produced by a net force. The net force and the acceleration must be in the same direction.

If applied to the uniform circular motion, the acceleration becomes the centripetal acceleration and the net force. Since it is also directed towards the axis of rotation, becomes the centripetal force.

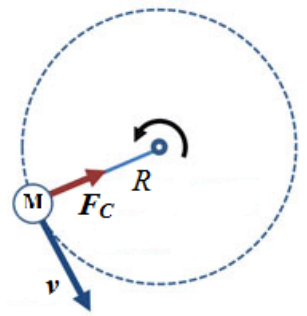
CENTRIPETAL FORCE can be defined as the net force on an object that maintains the object circular path and it is directed towards the center of the circular path.

$$F_c = Ma_c :$$

THEORY

When an object of mass M is revolving in a circular motion of radius R , the object is in accelerating motion. The radial component of the acceleration, called centripetal acceleration is given by which is directed to the center of the circular orbit. In a uniform circular motion, the speed, v , of the velocity vector is constant. Only direction is changing and the velocity is tangential to the orbit. So, the net force, called centripetal force, is also directed towards the center and given by

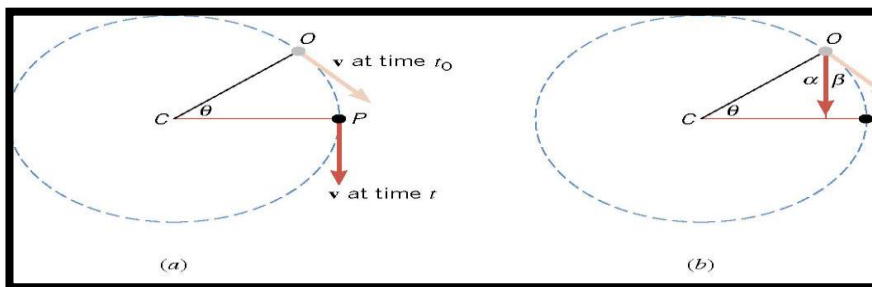
$$a_c = \frac{v^2}{R},$$



A CENTRIPETAL FORCE is not an extra force that occurs by itself. It is the resultant of some other forces such as tension, gravity, friction, elasticity, electric attraction etc. that cause the object to move in a circular path.

$$F_c = Ma_c = \frac{Mv^2}{R},$$

According to the Equation (2), centripetal force is proportional to the square of the speed for an object of given mass M rotating in a given radius R . You are going to experimentally verify this relationship in this lab. Similarly, you can investigate relation between any two quantities experimentally by keep two other quantities constant.



ROTATIONAL KINEMATICS

KINEMATICS AND NEWTON'S SECOND LAW OF MOTION provided by the basis for analyzing motion from one place to another and the forces that cause the motion to change.

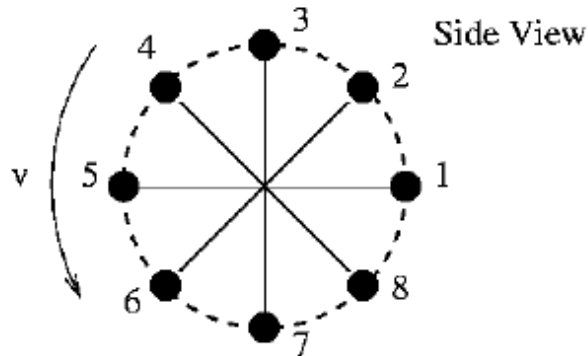
Axis objects tend to rotate on an imaginary straight line in the middle.

ROTATION object turns about an external axis or the axis within the object.

REVOLUTION the motion in an objects turns about an external axis.

SPEED is defined as distance traveled in a given time interval. Applying this definition, it is clearly seen that the horse on the outermost part will cover a large distance than the ones in a middle.

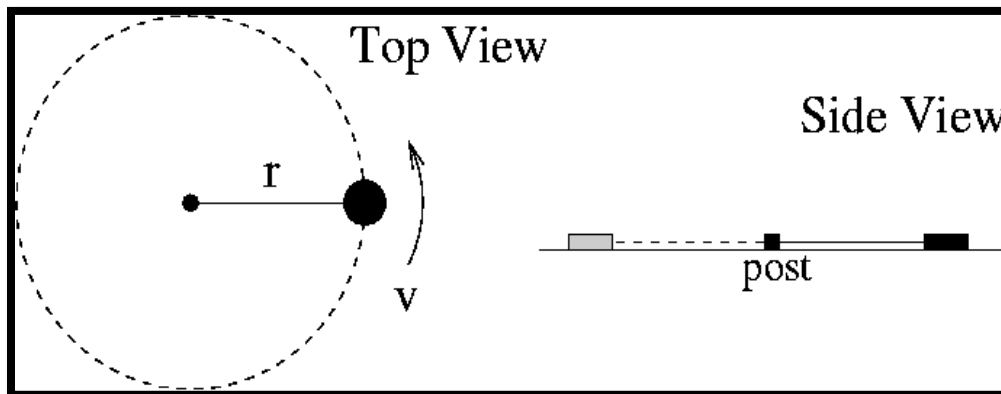
TANGENTIAL SPEED is the motion that we always tangent to the circle. Another name of tangential speed is instantaneous linear speed at that particular point in the circle.



ROTATIONAL SPEED is also another speed involved in merry-go-round that does not consider distance. This speed counts the number of rotations done in a given time interval.

EXAMPLES

1. A puck is traveling at a constant speed around a circle on a table. What would we have to know to find the tension in the sting



TENSION is the centripetal force (keeps motion in a circular path).

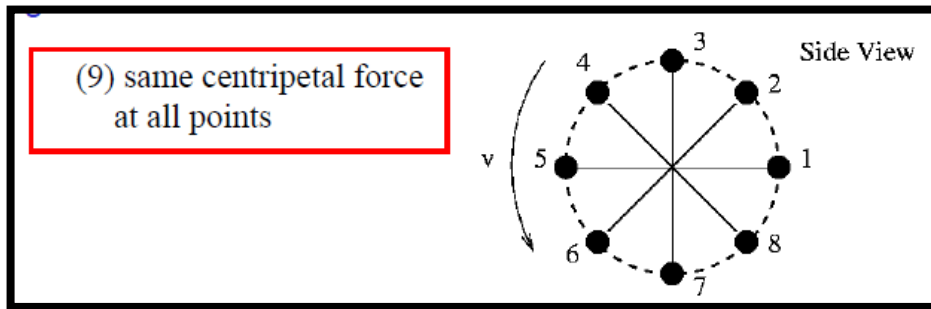
CENTRIPETAL ACCELERATION

$$a_c = v^2 / r$$

MASS OF THE BODY

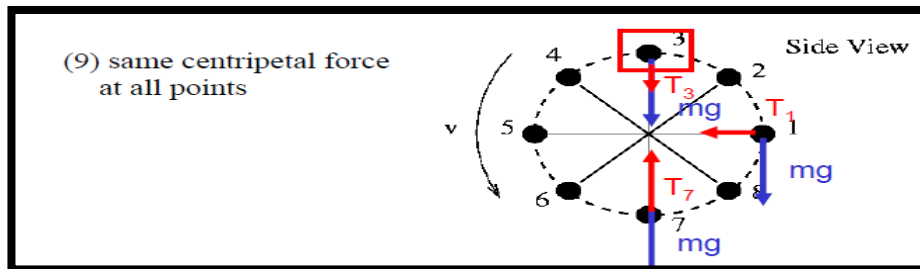
$$(F_c = m a_c)$$

You are twirling a tennis ball on a string in the vertical direction at a *constant* speed. At what point is the centripetal force the greatest?



Both gravity and tension have components in the radial direction. Same centripetal force (net force) at all points!

In 3, gravity works with the tension to keep it in circle:



$F_c = T_3 + mg$ (tension force does not have to be as large). If speed is truly constant, then $F_{centri} = m(v^2/r)$ and is the same for all points.

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ACTIVITY 1:

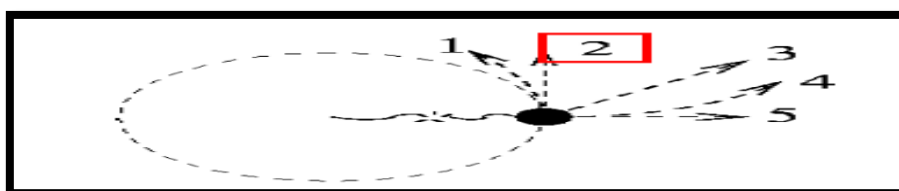
SOLVING PROBLEMS

PROCEDURE

1. Solving the problems involving Newton's laws of motion.
2. Study the characteristics of uniform circular motion
3. Experimentally measure centripetal force in a circular motion.

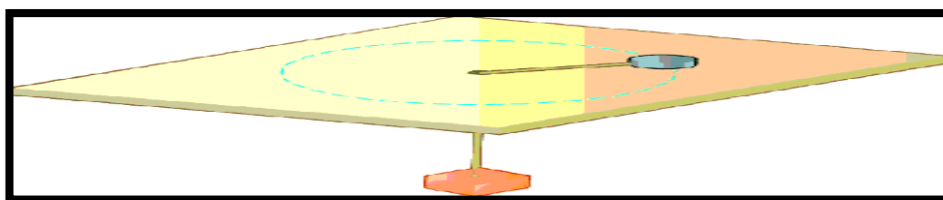
PROBLEMS

1. You are whirling a tennis ball on a string around in circles when the string suddenly snaps. What direction does the tennis ball fly? (the figure below is a top view)



- Instant velocity at any point is *tangent* to the circle.
- String snaps=no force acting on the puck.
- Newton's 1st Law: puck will move at constant speed on a straight line.

A puck of mass 0.25kg is tied to a string and allow to revolve in a circle of radius 1.0m on a Friction less table top. The other end of the string passes through a hole in the center of the table, and a mass of 0.5kg is tied to it. The suspended mass remains in equilibrium while the puck on the table rerevolves.



QUESTIONS

1. What is the tension in the string?
2. What is the force causing the centripetal acceleration of the puck?
3. What is the speed of the puck?

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ACTIVITY 1:

SOLVING PROBLEMS

1. . A car traveling at a given speed, v , can just barely make it around a bend of radius R without skidding sideways. Under identical road conditions, if this person wants to double their speed, the tightest turn they can make without skidding is a turn of radius:
 - (1) $(1/4)R$ (2) $\sqrt{1/2}R$ (3) $(1/2)R$
 - (4) R (5) $2R$ (6) $\sqrt{2}R$
 - (7) $4R$



Static friction between tires and road keeps car in circular motion of radius R . Identical road conditions, same friction.

Assuming same force, then same a_c : If double speed, must quadruple radius to keep a_c constant.

$$a_c = \frac{v^2}{R},$$

If double speed, must quadruple radius to keep a_c constant.

Both gravity and tension have components in the radial direction. Same centripetal force (net force AT all points



TEST YOURSELF!

PROCEDURE

1. Solving the problems involving Newton's laws of motion.
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1. A pilot of mass 70kg pulls out of a dive following a circular path of radius R at a speed of V. If we consider the pilot at the bottom of the dive (the plane is momentarily level), under which condition does the pilot feel the greatest acceleration? (calculate this acceleration)

(1) R=1800m, V=125 m/s

(2) R=900m, V=125 m/s

(3) R=450m, V=125 m/s

(4) R=1800, V=250 m/s

Centripetal acceleration: largest V, smallest R

$$a_c = \frac{v^2}{R},$$

What is the normal force (apparent weight) of pilot in this case?

GLOSSARY OF TERMS

CENTRIPETAL FORCE can be defined as the net force on an object that maintains the object circular path and it is directed towards the center of the circular path.

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